



IT

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INTRODUCTION TO IT AND BUSINESS SYSTEMS

FUNDAMENTALS OF INFORMATION TECHNOLOGY

- IT is the study and application of computers, software, and networks to manage information.
- It enables data storage, processing, and communication.
- IT is used in almost every sector: education, healthcare, banking, and business.

COMPONENTS OF IT

- **Hardware:** physical devices (CPU, memory, printers).
- **Software:** programs that run on hardware (Windows, MS Word, apps).
- **Networks:** systems that connect computers (LAN, WAN, Internet).
- **Databases:** organized storage of information for easy access.

ROLE OF IT IN BUSINESS OPERATIONS

- Automates routine tasks like payroll, billing, and inventory.
- Improves communication through emails, video calls, and instant messaging.
- Provides managers with tools for data analysis and decision-making.

ROLE OF IT IN BUSINESS OPERATIONS

- Enables e-commerce platforms (Amazon, Flipkart).
- Reduces costs by streamlining processes.
- Supports remote work and global collaboration.

OVERVIEW OF BUSINESS SYSTEMS

- ERP (Enterprise Resource Planning): integrates finance, HR, supply chain, and production.
- CRM (Customer Relationship Management): manages customer data, sales, and service.
- SCM (Supply Chain Management): tracks goods, suppliers, and logistics.

ERP EXAMPLE

- ERP uses a centralized database shared across departments.
- Real-time updates keep sales, inventory, and finance in sync.
- Benefits: faster reporting, reduced duplication, better coordination.

CRM EXAMPLE

- CRM stores customer details, purchase history, and preferences.
- Helps companies personalize marketing and improve customer service.
- Builds loyalty and long-term relationships.

INTERPLAY BETWEEN IT & BUSINESS STRATEGY

- IT is not just support—it drives business strategy.
- Companies use IT to innovate, expand globally, and gain competitive advantage.
- Example: Amazon uses IT for logistics, personalization, and predictive analytics.

DATA TYPES, SOURCES, AND LIFECYCLE

UNDERSTANDING DIFFERENT DATA TYPES

- Structured: organized in rows/columns (databases, spreadsheets)
- Unstructured: free-form (text, images, videos)
- Semi-structured: mix of both (XML, JSON, emails)

Data comes in different forms. Structured data is highly organized, like student records in a database. Unstructured data includes things like social media posts, images, or videos that don't fit neatly into tables. Semi-structured data is in between—emails, XML, or JSON files have some structure but are not fully tabular. Understanding these types helps businesses decide how to store and analyze information.

COMMON DATA SOURCES

- Internal: company records, employee data, sales reports
- External: market research, social media, government data
- Databases: relational (SQL) or non-relational (NoSQL)
- APIs: connect systems and share data

Data can come from inside the organization (like HR records or sales reports) or outside (like customer feedback, competitor analysis, or government statistics). Databases store structured information, while APIs allow systems to exchange data automatically. Businesses rely on both internal and external sources for decision-making.

DATA COLLECTION METHODS

- Surveys & questionnaires
- Transaction records
- Sensors & IoT devices
- Web scraping & APIs

Data is collected in many ways. Surveys and questionnaires gather opinions. Transaction records capture sales and purchases. Sensors and IoT devices collect real-time data (like temperature or machine performance). Web scraping and APIs pull data from websites or applications. Choosing the right method depends on the type of data needed.

DATA LIFECYCLE MANAGEMENT

- Creation: data generated or collected
- Storage: saved in databases/cloud
- Usage: analyzed for insights
- Archiving: stored for long-term reference
- Deletion: removed when no longer needed

Data goes through a lifecycle. It is first created or collected, then stored securely. Businesses use it for analysis and decision-making. Later, it may be archived for compliance or historical records. Finally, data is deleted to free space or meet privacy regulations. Managing this lifecycle ensures efficiency and security.

DATA QUALITY AND INTEGRITY

- Accuracy: data must be correct
- Consistency: same across systems
- Completeness: no missing values
- Reliability: trustworthy and secure

High-quality data is essential. Accuracy means the data reflects reality. Consistency ensures the same data is used across all systems. Completeness avoids gaps that could mislead analysis. Reliability means the data is secure and can be trusted. Poor data quality leads to wrong decisions, so integrity is critical.



INTRODUCTION TO BUSINESS ANALYST ROLE

DEFINITION OF A BUSINESS ANALYST

- BA = link between business and IT
- Identifies needs, proposes solutions
- Ensures projects meet objectives

A Business Analyst is a professional who understands business problems and translates them into technical solutions. They act as a bridge between stakeholders and IT teams, ensuring that projects deliver value.

RESPONSIBILITIES OF A BUSINESS ANALYST

- Gather and document requirements
- Analyze processes and suggest improvements
- Support testing and implementation

BAs collect requirements from stakeholders, analyze workflows, and recommend improvements. They also help in testing solutions and ensuring smooth implementation.

KEY SKILLS: COMMUNICATION

- Clear verbal & written communication
- Active listening
- Negotiation with stakeholders

Strong communication is essential. BAs must explain technical ideas in simple terms, listen carefully to stakeholders, and negotiate between conflicting needs.

KEY SKILLS: ANALYTICAL THINKING

- Break down complex problems
- Identify patterns & trends
- Evaluate options logically

Analytical thinking allows BAs to understand complex business problems, find root causes, and propose practical solutions.

KEY SKILLS: PROBLEM-SOLVING

- Creative solutions to challenges
- Balance business & technical needs
- Ensure feasibility

Problem-solving is about finding workable solutions that satisfy both business goals and technical constraints.

BUSINESS ANALYSIS METHODOLOGIES

- Waterfall: step-by-step, linear approach
- Agile: iterative, flexible, customer-focused
- Choice depends on project type

Waterfall is structured and sequential, while Agile is adaptive and iterative. BAs must understand both to work in different project environments.

AGILE METHODOLOGY IN BA

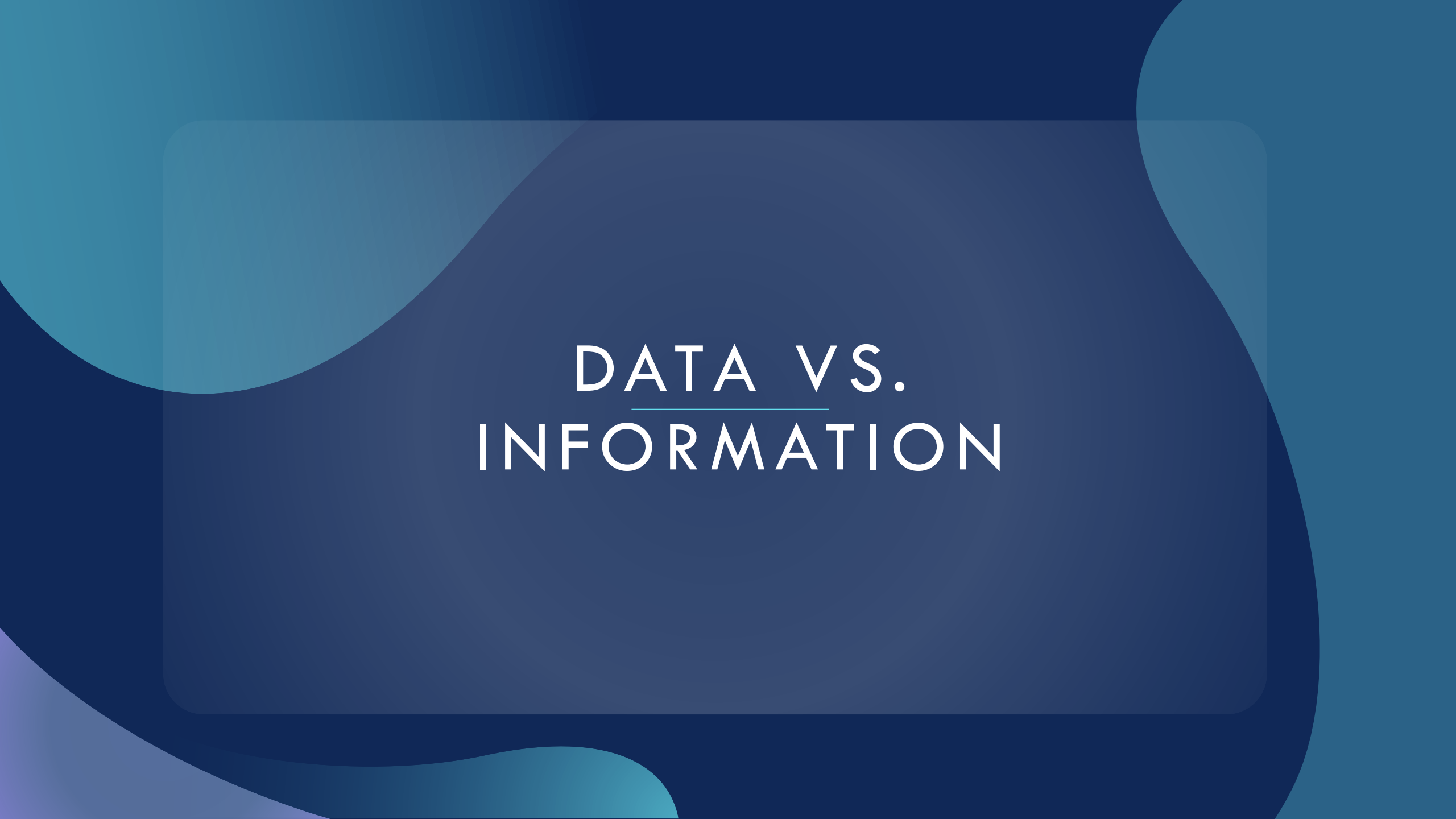
- Short sprints & continuous feedback
- BA works closely with product owner
- Focus on delivering value quickly

In Agile, BAs collaborate with teams in short cycles, refining requirements continuously and ensuring customer satisfaction.

STAKEHOLDER MANAGEMENT

- Identify stakeholders (internal & external)
- Manage expectations
- Build trust & collaboration

Stakeholder management is critical. BAs must balance different interests, keep communication open, and ensure stakeholders feel involved.



DATA VS. INFORMATION

INTRODUCTION TO DATA AND INFORMATION

- Data = raw facts, figures, symbols
- Information = processed, organized, meaningful data
- Foundation of knowledge and decisions

Data is the raw material—numbers, text, or observations. Information is what we get after processing and organizing data so it becomes useful. Example: "100, 95, 80" is data; "Average score = 91.6" is information.

RAW DATA CHARACTERISTICS

- Unprocessed, unorganized
- May be incomplete or scattered
- Examples: exam marks, sensor readings

Raw data is like a collection of facts without context. It doesn't tell us much until we process it. For instance, a list of temperatures is just data until we analyze trends.

PROCESSED INFORMATION CHARACTERISTICS

- Organized and structured
- Provides meaning and context
- Examples: reports, summaries, charts

Information is data that has been processed into a useful form. A sales report showing monthly totals is information derived from raw transaction data.

DISTINGUISHING DATA VS. INFORMATION

- Data: input, raw facts
- Information: output, meaningful insights
- Example: "500 units sold" → "Sales increased by 10%"

Data is the input, while information is the output. Businesses collect data but rely on information to make decisions.

TRANSFORMATION PROCESS

- Data → Information → Knowledge
- Processing adds meaning
- Knowledge = applied information

Data becomes information when processed. Information becomes knowledge when applied to solve problems or make decisions. Example: Data = sales numbers; Information = sales trend; Knowledge = strategy to increase sales.

DATA TO INFORMATION EXAMPLE

- Raw data: "Marks = 45, 60, 75"
- Information: "Average = 60, Pass rate = 100%"
- Knowledge: "Class performance is satisfactory"

This shows how raw marks (data) are processed into averages (information), which then helps teachers understand performance (knowledge).

VALUE OF INFORMATION IN DECISION MAKING

- Reduces uncertainty
- Supports planning and strategy
- Improves efficiency and accuracy

Information helps managers make better decisions. For example, analyzing customer feedback (information) guides product improvements.

BUSINESS EXAMPLE

- Data: customer purchase records
- Information: "Most sales from 18–25 age group"
- Knowledge: "Target marketing to young adults"

Businesses use data to generate insights. Information about customer demographics leads to knowledge that shapes marketing strategies.



OVERVIEW OF DATA-DRIVEN DECISION MAKING

INTRODUCTION TO DATA-DRIVEN DECISION MAKING

- Decisions based on facts, not guesswork
- Uses data analysis & insights
- Improves accuracy and confidence

Data-driven decision making means using evidence and analysis rather than intuition. It helps organizations make choices that are more reliable and measurable.

PRINCIPLES OF DATA-DRIVEN DECISION MAKING(PART1)

- Collect relevant, accurate data
- Ensure transparency in analysis
- Align data with business goals

Principles include gathering the right data, making analysis transparent, and ensuring that data supports organizational objectives.

PRINCIPLES OF DATA-DRIVEN DECISION MAKING (PART 2)

- Encourage evidence-based culture
- Continuous monitoring and improvement
- Ethical use of data

Organizations must build a culture where evidence guides decisions, monitor outcomes regularly, and use data responsibly.

BENEFITS OF USING DATA FOR DECISIONS (PART 1)

- Reduces uncertainty
- Improves efficiency
- Enhances accuracy

Data reduces risks by providing clarity, improves efficiency by streamlining processes, and ensures decisions are based on facts.

BENEFITS OF USING DATA FOR DECISIONS (PART 2)

- Supports innovation
- Strengthens customer focus
- Provides competitive advantage

Data helps companies innovate, understand customer needs better, and stay ahead of competitors.

STEPS IN THE DATA-DRIVEN DECISION PROCESS (PART 1)

- Identify problem or goal
- Collect relevant data
- Clean and prepare data

The process begins with defining the problem, gathering data, and preparing it for analysis.

STEPS IN THE DATA-DRIVEN DECISION PROCESS (PART 2)

- Analyze data for insights
- Make decision based on evidence
- Monitor and refine outcomes

Analysis generates insights, decisions are made based on those insights, and results are monitored for continuous improvement.

CHALLENGES AND PITFALLS

- Poor data quality
- Data overload (too much information)
- Resistance to change
- Misinterpretation of results

Challenges include inaccurate data, overwhelming amounts of information, organizational resistance, and the risk of drawing wrong conclusions.



CASE DISCUSSIONS / REAL-LIFE EXAMPLES

ANALYSIS OF BUSINESS SCENARIOS

- Identify the problem or challenge
- Collect and examine relevant data
- Generate insights and possible solutions
- Evaluate outcomes and refine decisions

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Business scenario analysis is about studying real situations to make better decisions. First, the problem or challenge is identified—for example, declining sales or supply chain delays. Next, relevant data is collected and examined to understand the root cause. Insights are then generated, leading to possible solutions such as new marketing strategies or system improvements. Finally, outcomes are evaluated, and decisions are refined to ensure long-term success. This process shows how IT and data systems support practical business problem-solving.



INTRODUCTION TO DATA GOVERNANCE & DATA ETHICS

WHAT IS DATA GOVERNANCE?

- Framework for managing data assets
- Ensures accuracy, security, and availability
- Aligns data with business goals

Data governance is the overall management of data within an organization. It ensures that data is reliable, secure, and used effectively to support business objectives.

POLICIES AND PROCEDURES

- Define rules for data usage
- Establish standards for quality & security
- Guide compliance with regulations

Policies and procedures set the rules for how data is collected, stored, and shared. They ensure consistency and protect against misuse.

ROLES AND RESPONSIBILITIES

- Data stewards: maintain quality
- IT teams: ensure security & infrastructure
- Business leaders: align data with strategy

Different roles are involved in governance. Data stewards focus on accuracy, IT teams handle security, and business leaders ensure data supports organizational goals.

PRINCIPLES OF DATA ETHICS

- Fairness in data use
- Respect for individuals
- Responsible handling of information

Data ethics ensures that data is used in ways that respect people's rights and avoid harm. It emphasizes fairness and responsibility.

PRIVACY AND CONFIDENTIALITY

- Protect personal information
- Limit access to sensitive data
- Follow consent and legal requirements

Privacy means safeguarding personal data. Confidentiality ensures only authorized people can access sensitive information, protecting individuals and organizations.

TRANSPARENCY AND ACCOUNTABILITY

- Clear policies on data use
- Explain how data is collected & shared
- Hold organizations accountable

Transparency builds trust by showing how data is handled. Accountability means organizations take responsibility for misuse or breaches.

REGULATORY COMPLIANCE

- GDPR (Europe): protects personal data
- CCPA (California): consumer privacy rights
- Organizations must follow laws globally

Regulations like GDPR and CCPA set strict rules for data protection. Companies must comply to avoid penalties and maintain trust.

BUILDING A CULTURE OF RESPONSIBLE DATA USE

- Train employees on ethics & governance
- Encourage responsible practices
- Promote trust and integrity

Responsible data use requires a cultural shift. Employees must be trained, and organizations should encourage ethical practices to build trust.



THANK YOU

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